

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
CO1_AT3 — VISUALIZATION DESIGN EXERCISE

Course Code:	DSA0606	Course Title:	Data Handling and Visualization
CO Assessed:	CO1 – Precision (BL2)	Assessment:	Assessment Tool 3 – Visualization Design
Weightage:	40% of CIA	Total Marks:	20 (2 Questions × 10 Marks)
CO1 Statement:	<i>Understand the fundamental concepts, principles, and techniques of data visualization.(BL2)</i>		
Student Name:	M.SAI HARSHITHA	Reg. No.:	192424408
Section / Batch:	B.TECH AI & DS	Date:	27.07.2026

Code of Conduct

I M.SAI HARSHITHA/192424408 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: M.SAI HARSHITHA

Instructions

- This assessment contains 2 questions, each carrying 10 marks. Total: 20 Marks.
- No negative marking. Unanswered questions carry zero marks.
- No calculators or electronic devices permitted.
- Duration: 60 minutes.

Annexure A – VISUALIZATION DESIGN

ASSESSMENT RUBRIC

AT3 — Visualization Design Exercise

CO1 · BT3: Apply ·

This rubric is used to assess student responses to the AT3 Visualization Design Exercise problems, in which students design and justify a complete visualization (chart type, aesthetic mappings, scales, coordinate systems, and/or color scheme) for a given dataset or design brief.

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of chart types, aesthetic mappings, scales, coordinate systems, and color scales, and correctly identifies which apply to the given design scenario.	Good understanding of the relevant design concepts, with only minor gaps in selecting appropriate chart types, aesthetic mappings, or scales.	Basic understanding of design concepts, but with some misconceptions about appropriate chart types, aesthetics, or scales for the scenario.	Poor understanding of core design concepts; proposes chart types or mappings that are fundamentally unsuited to the data.
Explanation	25%	Provides detailed, well-reasoned justification for every design decision, explicitly connecting each choice to the properties of the data and the scenario.	Provides clear justification for design decisions, with adequate reasoning for most sub-questions.	Justification is limited or superficial; design choices are stated without sufficient reasoning.	Little or no justification provided; design choices appear arbitrary or unexplained.
Application	20%	Effectively applies visualization design principles to produce a complete, appropriate, and well-reasoned design solution for the specific scenario.	Applies design principles to produce an appropriate solution, with only minor errors or missed refinements.	Limited application of design principles; the proposed design partially addresses the scenario but has notable weaknesses.	Fails to apply appropriate design principles; the proposed design does not suit the scenario or data.
Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning

Organization	15%	The design proposal is clearly structured, addressing each sub-question (a, b, c) in a logical sequence with coherent flow throughout.	The proposal is organized and addresses each sub-question, with only minor issues in flow or clarity.	Basic structure; sub-questions are addressed but the proposal lacks clarity or a logical sequence.	Poorly organized; the proposal is incoherent, and sub-questions are left unaddressed or answered out of order.
Accuracy	10%	All design choices — chart type, aesthetic mapping, scale, coordinate system, and/or color scheme — are completely accurate and well-suited to the data.	Mostly accurate design choices, with only minor omissions or a small error in one part of the proposal.	Partially correct; some design elements are missing, mismatched to the data, or incorrect.	Design choices are largely incorrect or inappropriate, with major gaps in matching the visualization to the data.

Scoring Guide

For each criterion, award a performance level (1–4) based on the descriptors above. Multiply the level achieved (as a percentage of 4, i.e., Level 4 = 100%, Level 3 = 75%, Level 2 = 50%, Level 1 = 25%) by the criterion's weightage, then sum across all five criteria to obtain the total score out of 100%.

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

AT3 — Visualization Design Exercise

NAME: M.SAI HARSHITHA

REG.NO:192424408

COURSE CODE: DSA0606

COURSE NAME:DATA HANDLING AND VISUALIZATION

Problem 6 (10 Marks)

Scenario: A school wants to show the number of students per grade level (1st through 12th) alongside average test scores per grade.

- Design a chart (or combination of charts) to show both variables. Explain the aesthetic mapping decisions.
- Would you use a single dual-axis chart or two separate charts? Justify your choice, including any risks of a dual-axis design.
- What data type is “grade level,” and how does that influence your choice of x-axis ordering?

DATA VISUALIZATION AND CHART DESIGN

Dataset:

Grade Level	Number of Students	Average Test Score
1st	120	82
2nd	115	85
3rd	130	78
4th	125	88
5th	140	84
6th	135	80
7th	150	86
8th	145	89
9th	160	83
10th	155	91
11th	145	87
12th	135	92

a) Design a chart to show both variables

Chart Type: Combination Chart (Bar Chart + Line Chart)

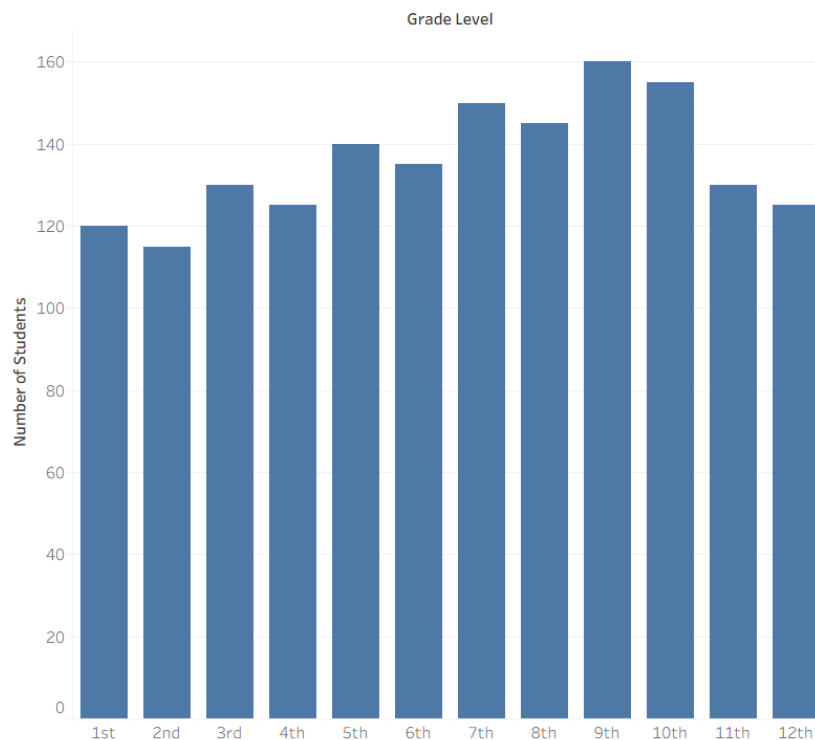
- A combination chart is used to represent both variables together.
- The bar chart represents the number of students because student count is a quantitative value, and bars make comparison between different grade levels easier.
- The line chart represents the average test score because it helps identify trends, improvements, and variations in performance across grades.

Chart Mapping:

- X-axis: Grade Level
- Bar Height: Number of Students
- Line Position: Average Test Score
- Bar Color: Student Count
- Line Marker: Average Score

Visualization Decisions:

- Grade level is placed on the X-axis because it is the category variable.
- Student count is represented using bar height.
- Average test score is represented using line position.
- Different colors are used to distinguish student count and average score clearly.
- Data labels can be added to improve readability.



b) Single Dual-Axis Chart or Two Separate Charts?

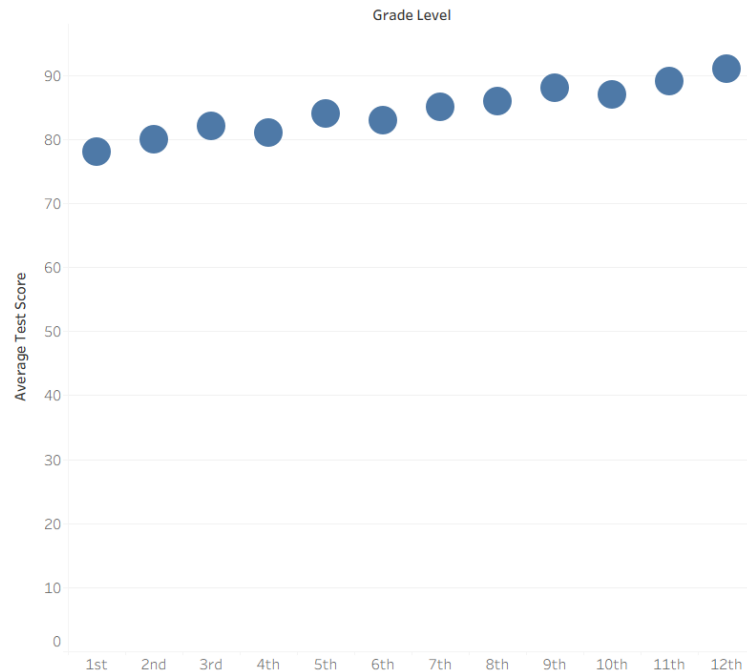
Recommended Choice: Two Separate Charts

Two separate charts are preferred because student count and average test score are different types of measurements with different units.

A dual-axis chart may create misleading interpretations because different axis scales can change how the relationship between variables appears.

Two separate charts provide:

- Better comparison
- Clear visualization
- Accurate interpretation



c) Data Type of Grade Level

Data Type: Ordinal Data Type

Grade level is an ordinal data type because it has a meaningful order.

Example:

1st → 2nd → 3rd → ... → 12th

However, the difference between grade levels is not a numerical measurement.

Effect on Visualization:

Since grade level is ordinal data, the X-axis should maintain the natural order.

Correct Order:

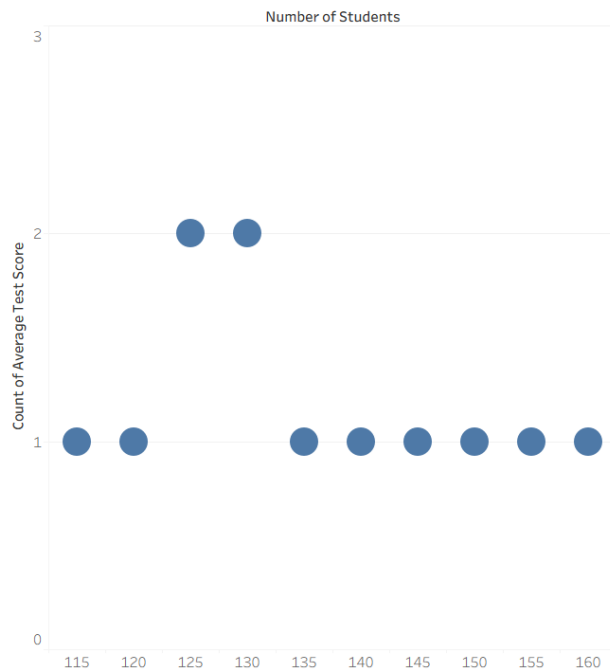
1st, 2nd, 3rd, 4th, 5th, 6th, 7th, 8th, 9th, 10th, 11th, 12th

Incorrect Order:

1st, 10th, 11th, 12th, 2nd, 3rd

Conclusion:

A combination chart is suitable for visualizing student count and average test scores. Grade level should be treated as ordinal data and displayed in its natural sequence for accurate visualization.



Problem 16 (10 Marks)

Scenario: A geography teacher wants to create a map showing average annual rainfall across 15 districts, ranging from very low to very high rainfall.

- a) Design a choropleth map for this data. Which type of color scale would you choose, and why is it appropriate here?
- b) Specify the two end colors of your scale and explain your reasoning (e.g., avoiding colorblind-unfriendly combinations).
- c) How would you handle a district with missing rainfall data in your color design?

Problem 16: Choropleth Map Design**Dataset:**

District	Average Annual Rainfall (mm)
District A	450
District B	520
District C	680
District D	750
District E	820
District F	950
District G	1100
District H	1250
District I	1400
District J	1550
District K	1700
District L	1900
District M	2200
District N	2500
District O	3000

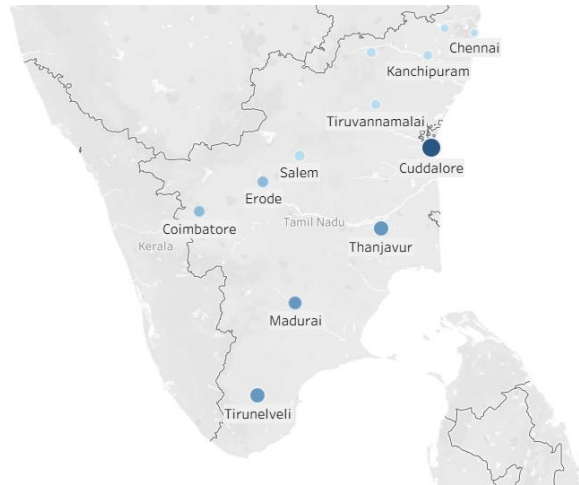
a) Design a Choropleth Map**Chart Type: Filled Map (Choropleth Map)**

A choropleth map is used to represent average annual rainfall across different districts by using colors to show variations in rainfall values.

Color Scale Used: Sequential Color Scale

Reason:

A sequential color scale is appropriate because rainfall values have a continuous range from very low to very high. Light colors represent areas with low rainfall, and darker colors represent areas with high rainfall. This makes it easier to identify rainfall patterns across districts.

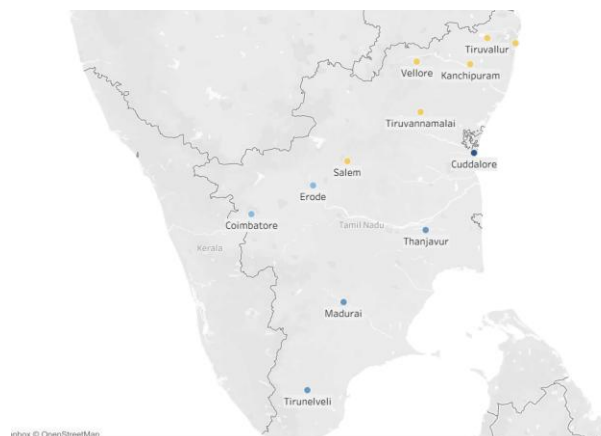


b) End Colors of the Scale

- Color Scale:
- Low Rainfall: Light Yellow
- High Rainfall: Dark Blue

Reasoning:

- Light yellow represents districts with very low rainfall, while dark blue represents districts with very high rainfall. Blue is associated with water and rainfall, making it easy to understand. The color combination avoids red-green combinations, which may cause difficulty for colorblind users.



c) Handling Missing Rainfall Data

If a district has missing rainfall data, it should be displayed using a neutral color such as light gray.

Reason:

A separate neutral color prevents users from interpreting missing data as either low or high rainfall. The map legend should include a label such as "Missing Data" for clarity.

Tableau Visualization:

Create a Filled Map in Tableau.

Steps:

1. Import the rainfall dataset into Tableau.
2. Select District as the geographic field.
3. Choose Filled Map from the Show Me option.
4. Drag Average Annual Rainfall to the Color section.
5. Apply a sequential color scale from light yellow to dark blue.

The final output will be a choropleth map showing rainfall variation across districts.

